Homework 8 (Python Practice)

March 18, 2026

This homework has a few exercises intended to help with learning/reviewing Python. Use any combination of your math knowledge (with appropriate proofs) and code to find the quantities below. Please submit your code as a .ipynb file or screenshots of all code used to solve the problems. You are also welcome to screenshot the output of the code (rather than typesetting your solution) should you wish. You are welcome (and encouraged) to use standard libraries such as the numpy and math libraries. Each problem is worth 10 points.

1)

Let $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for all $n > 1$. These are the Fibonacci numbers. Using some combination of code and/or mathematical reasoning, find the value of the following sum:

$$\sum_{k=0}^{100} (-1)^k F_k$$

2)

Let $A_{\text{Pow}} \in \mathbb{R}^{n \times n}$ be defined as

$$[A_{\text{Pow}}]_{ij} = j^{i-1}$$

for $1 \leq i, j \leq n$. For example, for $n = 3$, we have

$$A_{\text{Pow}} = \begin{pmatrix} 1^0 & 2^0 & 3^0 \\ 1^1 & 2^1 & 3^1 \\ 1^2 & 2^2 & 3^2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

Find the largest entry of A_{Pow}^{-1} when $n = 10$. Round your solution to 5 decimal places.

3)

Consider A_{Pow} as defined above. Find $\det(A_{\text{Pow}} - 3I)$ when $n = 6$. (Note: be careful of numerical error. The correct answer should be an integer).

4)

Let

$$A_{ij} = (i + j)!$$

Suppose $A_{ij} \in \mathbb{R}^{20 \times 20}$. Compute the sum of all the entries in A.

5)

Consider the set $S = \left\{1, 2, \ldots, (10)^5\right\}$. Determine how many elements k of S satisfy all of the following:

a) k is divisible by 7
b) The last digit of k is a 3
c) The sum of the digits of k is less than 20

6)

The triangle numbers are defined recursively as $T_0 = 0$ and $T_n = T_{n-1} + n$ for all $n \geq 1$. Let $S = \left\{T_n | 1 \leq n \leq (10)^5\right\}$. Determine the number of triangle numbers in S that are divisible by 4.